

Capacitance Level Switch For Solid Application

Model: CPLS-900S



PRODUCT DESCRIPTION:

The CPLT-900S Capacitance Level Switch is a rugged, reliable and cost-effective point level switch designed for level detection of bulk materials in bins, hoppers and silos. It is suitable for High Level, Low Level and Differential Level detection and can be used across a wide range of industries including cement, material handling, chemical, power, food, pulp & paper, pharmaceutical and foundry applications. The probe is designed to minimize the effect of material build-up, providing reliable level detection in applications where the material tends to stick or accumulate on the probe. The unit incorporates integral electronics with adjustable time-delay, LED indication and Fail-Safe operation, ensuring dependable and convenient level monitoring and control.

SALIENT FEATURES:

- Wide range of industrial applications
- Easy installation and maintenance
- Weatherproof / Flameproof enclosure as per application requirement
- PTFE-insulated electrode suitable for temperatures up to 180°C
- No moving parts – no mechanical wear and tear
- Adjustable differential time delay with ON/OFF timing
- Field-configurable Fail-Safe High / Low selection
- Reliable detection despite sticky and conductive material build-up between the probe and vessel wall

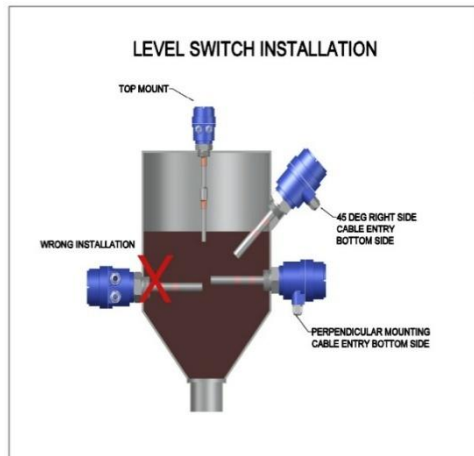
OPERATING PRINCIPLE:

The CPLT-900S Capacitance Level Switch operates on the principle of change in capacitance between the sensing probe and the vessel wall. The probe and the vessel form a capacitance whose value changes when the process material comes into contact with the probe. As the material level rises and covers the probe, the dielectric constant between the probe and the vessel changes, resulting in a corresponding change in capacitance. The integral electronic circuit detects this change and generates a switching signal to operate the output relay.

The instrument incorporates build-up compensation, which helps minimize the effect of material coating on the probe and provides reliable level detection even in applications involving sticky or conductive materials.

Capacitance Level Switch For Solid Application Model: CPLS-900S

PROBE CONSTRUCTION SUITABLE FOR MATERIAL BUILD-UP:



The CPLT-900S is available with a specially designed COAT Guard electrode to ensure reliable level detection in applications where material build-up may occur on the probe.

A conventional probe may be affected by material coating or build-up, which can result in false level signals. The COAT Guard uses a dedicated build-up compensation technique to minimize the effect of material accumulated on the probe. The COAT Guard electrode is electrically insulated from the active sensing electrode and is driven with the same RF frequency and polarity. When material builds

up on the probe, the COAT Guard circuitry creates a saturated sensing zone around the build-up area, preventing the RF signal from flowing through the build-up to the vessel wall. When the actual material level rises and reaches the sensing probe, the RF signal establishes a path from the active sensing electrode to ground, beyond the compensated build-up region. The electronics then detects this change and provides a reliable level switching signal. This construction makes the probe particularly suitable for sticky, coating and material build-up applications.

THE PART OF COMPACT LEVEL SWITCH ARE EXPLAINED IN FIGURE

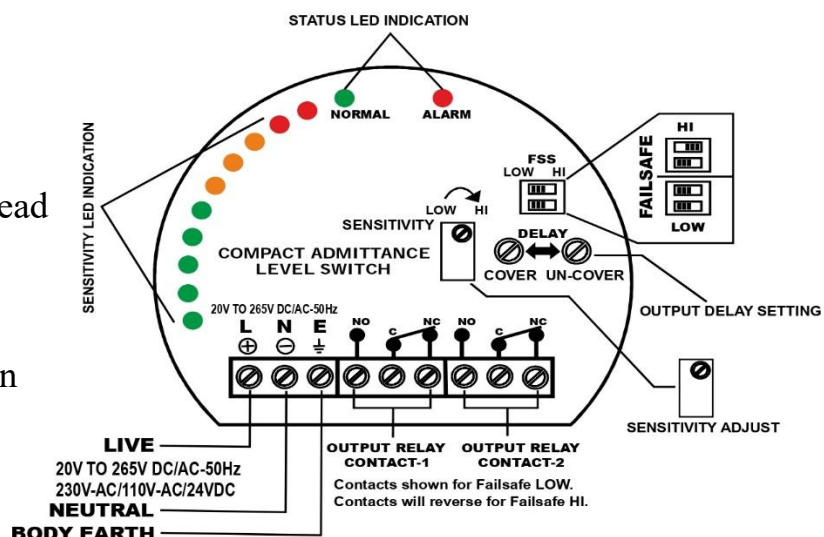


- ---Aluminum Housing
- -----Cable Entry
- -----Mounting Thread
- -----PTFE insulation
- -----sense

ELECTRICAL CONNECTION

Universal /230 VAC/110VAC/24 VDC

ELECTRICAL CONNECTION- UNIVERSAL



Capacitance Level Switch For Solid Application

Model: CPLS-900S

DATA SHEET

CPLS-900– Capacitance Level Switch

Selection / Ordering Information

Selection	Code	Description
Model No.	CPLT-900S	Capacitance Level Switch
Application	S	Solids
Mounting Type	IN	Integral
Probe Type	RD / RP/HD	Rigid Rod / Rope Flexible/Heavy Duty type Probe
Insulation	FP/PT/ CE	Full PTFE / Part PTFE / Ceramic
Probe Diameter	D1 / D2	Standard 16mm / Heavy Duty 30mm
Probe Material	04/ 16/ 6L/MS/ SM	SS304/ SS316/ SS316L /Mild steel/ Special Material
Power Supply	AC/DC/A1/UN	230VAC / 24VDC / 110VAC / Universal
Output	DP/SP/PN	DPDT Relay /SPDT Relay / PNP Relay
Switching Delay	ND / DA	No Delay / Adjustable
Indication	L	LED
Enclosure	WP / FL	Weatherproof / Flameproof
Connection	T	Threaded
	F	Flanged
Mounting	B1 / B2 / B3/B4	1" / 1½" / 2" BSP/ Special Mounting
	N1 / N2 / N3/N4	1" / 1½" / 2" NPT/ Special Mounting
Flange Rating	A1 / A3 / A6	ANSI 150# / ANSI 300# /ANSI 600#
Flange Facing	RF / FF	Raised Face / Flat Face
Flange Material	S4 / S6/ S16/MS/SL	SS304 / SS316 /SS316L /Mild steel / Special material
Flange Size	01 / 15 / 02 / 03 / 04/PS	1" / 1½" / 2" / 3" / 4"/ Special Process
Pressure	P1 /P2/P3/P4	Standard 0 to 2KG/ 2 To 10kg /10kg to 40kg / Customised
Temperature	T1/ T2/TS	Max -20–150°C / 200°C/ Customer Specified Special Designed
High Temperature stand off	H8/HS	80MM/Special standoff Extension
Cable Gland	G1 / G2 / G3	PG13.5 / ½"NPT / ¾" ET
Ground / Reference	GS / GX	Standard / Special
Material Certificate	M0 / MC	Not Required / Material test certificate

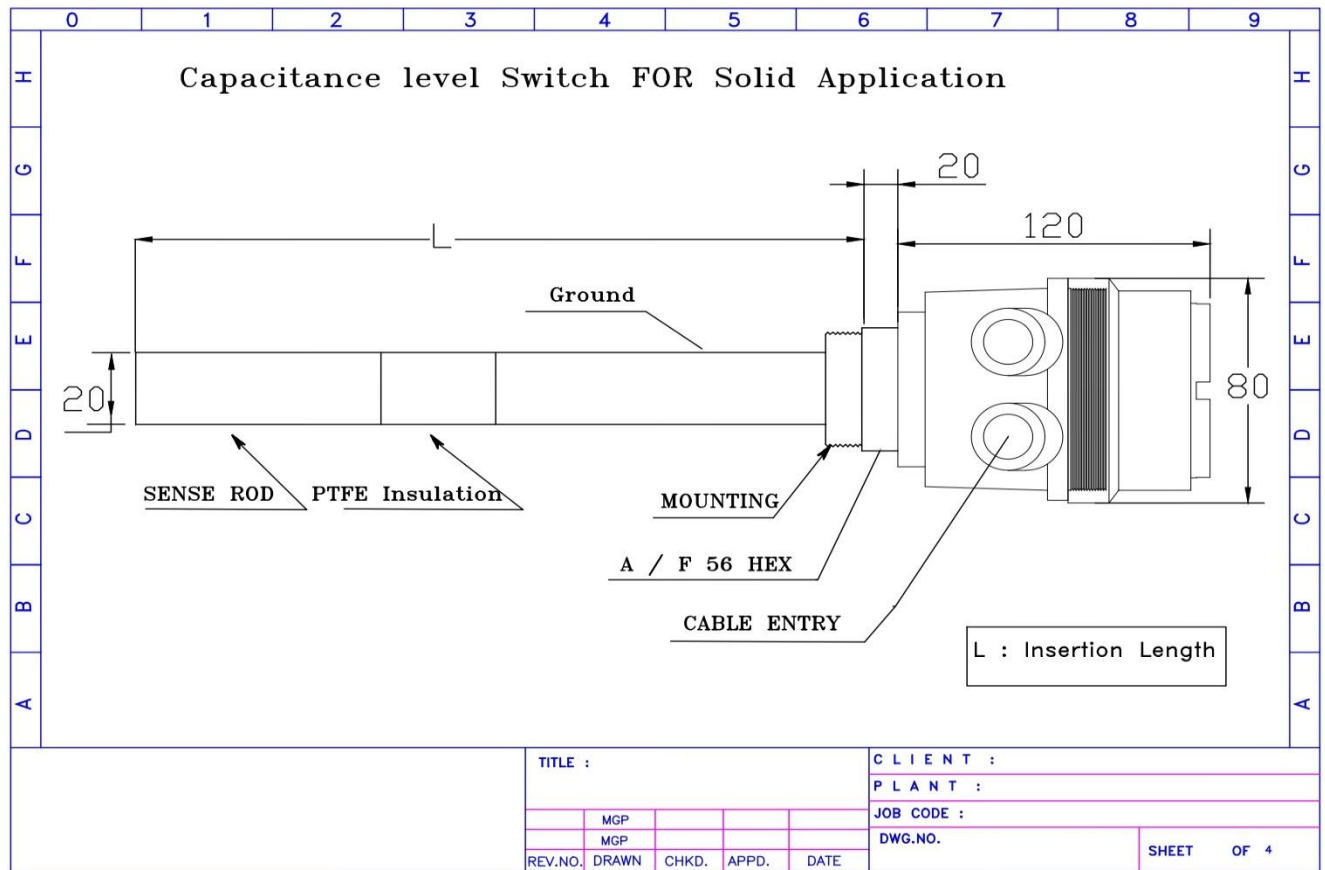


Capacitance Level Switch For Solid Application

Model: CPLS-900S

Selection	Code	Description
Test Certificate	TC / CC	Test Certificate / Calibration Certificate
Documentation	DS / GA / IR	Datasheet / GA / Inspection Report
Tagging	TAG	Customer Tag No.
Probe Length	XXXX	200mm To Customize

MECHANICAL DRAWING: INTEGRAL TYPE:



Manufactured By:

Levtron Instruments Pvt Ltd.

Office No 2, 1st floor, plot no 6, Pandhari Industrial estate, Shivane. Pune 411023, Contact: +91 9579430012/+91 9730042725,

E-mail: sales@levtron.in www.Levtron.in/
www.Levtroninstruments.com/ www.levelsence.in